

2024



Accelerating Car Design in a Sustainable and Flexible Environment Using HPC on AWS with TGR-E

Learn how TOYOTA GAZOO Racing Europe is optimizing its HPC simulation workloads using HPC on AWS.

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10% increase
in processing speed

95% re
in carbon f



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Motorsport vehicles need to withstand intense conditions like high speeds, tight turns, and impacts from crashes, and advanced simulations help companies like [TOYOTA GAZOO Racing Europe](#) (TGR-E) design cars that can handle these challenges. High performance computing (HPC) simulations are common in both the motorsport and automotive industries, but they require thousands of compute instances to perform a simulation and generate results. In an on-premises environment, this can create a bottleneck; teams of engineers running workloads with different characteristics must compete for a fixed and limited amount of compute nodes.

To overcome these capacity challenges and improve efficiency, TGR-E migrated its HPC workloads to Amazon Web Services (AWS). On AWS, the company can process complex workloads faster and at a lower cost than before. These enhanced simulation capabilities have helped accelerate motorsport design, contributing to faster, more efficient, and better cars.

TOYOTA GAZOO Racing accelerates design utilizing HPC and simulations on AWS

Opportunity | Using AWS to Support HPC Workloads for TGR-E

Based in Cologne, Germany, TGR-E is a cutting-edge engineering center and subsidiary of [Toyota Motor Corporation](#) (Toyota). It focuses on motorsport research and development for World Endurance Championship and World Rally Championship races, including the 24 Hours of Le Mans. Previously, the company ran its simulations using an on-premises HPC cluster, using only 60 percent of its resources on average but paying for 100 percent. It also encountered insufficient compute capacity during peak periods, especially before large events, causing engineers to wait for compute resources and delaying engineering results. With the on-premises cluster lease set to expire, TGR-E turned to AWS to run its HPC workloads.

“The idea was to follow the enterprise strategy of Toyota, find a solution in the cloud, and pay for only what we use,” says Oliver Meis, senior manager at TGR-E. “Although we averaged 60 percent usage for our compute resources, we still had peaks where even 100 percent capacity was not enough. By migrating to AWS, we were able to scale to cover these peaks as well.”



To support its migration from on premises to the cloud, TGR-E participated in the [AWS Migration Acceleration Program](#) (AWS MAP), a comprehensive and proven cloud migration program based on the AWS team's experience migrating thousands of enterprise customers to the cloud. The company completed AWS MAP working alongside [Eviden](#), an Atos business and [AWS Partner](#) that offers IT services and solutions for the efficient use of complex computer environments in research, development, and computation.

Solution | Achieving 10% Faster Simulation Run Times While Reducing Costs and Downtime

After testing different instance types, TGR-E migrated its on-premises HPC cluster to the HPC Optimized instance family of [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. These instances include Amazon EC2 [Hpc6a Instances](#), which are powered by third-generation AMD EPYC processors; Amazon EC2 [Hpc7a Instances](#), which are powered by fourth-generation AMD EPYC processors; and Amazon EC2 [Hpc6id Instances](#), which are powered by third-generation Intel Xeon Scalable processors. By adopting HPC Optimized instances, TGR-E gained the computational power required to run complex simulations efficiently at scale. It uses HPC on AWS to run multiple computer-aided engineering simulations using a variety of open-source and commercial packages in computational fluid dynamics, dynamic load analysis, finite element modeling, and more.

TGR-E created instantly deployable HPC architectures for application software with remote desktops based on [NICE DCV](#), a high-performance remote display protocol. This approach combines all open-source and commercial software accessed by engineering teams in a single environment on AWS. These simulations help engineers understand the behavior and interaction of liquids and gases with surfaces, which is crucial for optimizing the aerodynamics, cooling, and overall performance of motorsport vehicles.

TGR-E also adopted [AWS ParallelCluster](#), an open-source cluster management tool that simplifies the deployment and management of HPC clusters. Its engineers use this tool to automate the provisioning of resources for HPC applications; they submit simulation parameters to the job scheduler, and AWS ParallelCluster handles the rest, including the scaling up and down of Amazon EC2 compute nodes.

The solution accelerates HPC jobs by attaching Amazon EC2 instances to [Elastic Fabric Adapter](#) (EFA), a network interface for Amazon EC2 instances that provides low network latency and 300 Gbps of network bandwidth. As a result, MPI jobs using EFA have a lower latency compared to the traditional on-premises network without any additional cost.



To store and protect data, TGR-E uses [Amazon Simple Storage Service](#) (Amazon S3), an object storage service, alongside [Amazon FSx for Lustre](#), which provides fully managed shared storage with the scalability and performance of the popular Lustre file system. Amazon FSx for Lustre handles the company's most input- and output-intensive workloads. When linked to an Amazon S3 bucket, an Amazon FSx for Lustre file system presents Amazon S3 objects as files and lets engineers write results back to Amazon S3.

Since migrating to AWS, TGR-E has seen numerous benefits. From day one, the company has experienced a 10 percent increase in processing speed along with consistent and reliable performance. It has experienced no downtime or job slowdowns, leading to more efficient and robust workflows. Additionally, the cost benefits have exceeded expectations, with TGR-E unlocking significant savings by omitting traditional data center costs, such as rack space, cooling, and energy.

Outcome | Designing Motorsport Vehicles Faster, More Efficiently, and at a Reduced Cost

By taking advantage of HPC on AWS, TGR-E is accelerating motorsport car design to create better vehicles. TGR-E has also reduced the carbon footprint of its HPC workloads by 95% compared with on premises based on the AWS customer carbon footprint tool, which uses simple-to-understand data visualizations to help customers review, evaluate, and forecast emissions. TGR-E plans to continue its digital transformation, focusing on efficiency as a way to reduce emissions. By using advanced AWS technology, TGR-E optimizes its HPC applications while supporting the company's long-term strategic vision to develop carbon neutral mobility solutions. The process creates a flywheel effect as TGR-E continually brings new simulation workloads onto AWS and benefits from new Amazon EC2 instances to optimize current workflows. To optimize its HPC workloads even further, the company plans to explore [AWS Graviton processors](#), which are designed to deliver the best price performance for cloud workloads running on Amazon EC2.

"With HPC on AWS, we have much more flexibility to react to future demands or even surprise peaks," says Meis. "But the story won't end there. We now have the potential for continual improvement on AWS, and we can benefit from new technologies much sooner than we could on premises."

About TOYOTA GAZOO Racing Europe

TOYOTA GAZOO Racing Europe is an engineering center that focuses on automotive and motorsport research and development. Based in Cologne, Germany, it is the home of TOYOTA GAZOO Racing's FIA World Endurance Championship team.

AWS Services Used

Amazon EC2 Hpc6a Instances

Amazon Elastic Compute Cloud (Amazon EC2) Hpc6a instances feature 3rd Gen AMD EPYC processors and are designed for tightly coupled, compute-intensive high performance computing (HPC) workloads.



computational fluid dynamics (CFD), weather forecasting, and multiphysics simulations.

[Learn more »](#)

Amazon EC2 Hpc7a Instances

Amazon Elastic Compute Cloud (Amazon EC2) Hpc7a instances, powered by 4th Gen AMD EPYC processors, deliver up to 2.5x better performance compared to Amazon EC2 Hpc6a instances.

[Learn more »](#)

Amazon EC2 Hpc6id Instances

Amazon Elastic Compute Cloud (Amazon EC2) Hpc6id instances, powered by 3rd Generation Intel Xeon Scalable processors, offer cost-effective price performance for memory-bound and data-intensive high performance computing (HPC) workloads in Amazon EC2.

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NICE DCV

NICE DCV is a high-performance remote display protocol that provides customers with a secure way to deliver remote desktops and application streaming from any cloud or data center to any device, over varying network conditions.

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Continental on AWS

From one of the most recognized tire brands in the world to a connected mobility pioneer, Continental has seen success from 1871 to today thanks to its constant evolution. Using Amazon Web Services (AWS), the company has embraced the cloud and accelerated a software-driven vision. It built modular solution Continental Automotive Edge (CAEdge), which connects vehicles to



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ITALY



Ferrari on AWS

Luxury Italian auto manufacturer Ferrari S.p.A. (Ferrari) has forged a globally recognized legacy that's rooted in tradition and innovation over many decades. To continue to deliver the best possible experiences to customers and dealers, Ferrari has tapped into the power of generative artificial intelligence (AI) by turning to Amazon Web Services (AWS).

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